

When gold disconnects

Jack Pickett · 19th September 2026

Abstract

In the hire graph of the 3-free door, every odd prime other than 3 meets 5 in the infinite undirected gold graph. Finite windows can still be disconnected: each new 2-power-door prime is gold-isolated at its first-owner window. Downward gold chains terminate only at 2-power doors, so gold is disconnected for infinitely many finite X if and only if there are infinitely many Mersenne or Fermat primes. The companion note settles the infinite component of 5; this note records the equivalence for the leftover finite-window question, together with the thin M_{31} first-owner and first direct-bridge integers.

1 Setup

We use the notation of the hire-graph note [1]: the 3-free door m_0 , the hire set S_X , directed gold arcs $q \mid m_0(p)$, and the undirected gold graph on $S_X \setminus \{2\}$. A prime $M \neq 3$ is a *2-power door* if $m_0(M) = 2^k$ for some $k \geq 1$. Every Mersenne prime $M_p = 2^p - 1$ with p odd, and every Fermat prime greater than 3, is of this form. The first owner r_0 of M is the least prime r with $M \mid m_0(r)$; by the door bound, $r_0 \geq 2M - 1$.

2 First-owner windows

Lemma 1. *Let $M > 5$ be a 2-power-door prime and let r_0 be its first owner. At the window $X = r_0$, the prime M is an isolated vertex of undirected gold. In particular gold is disconnected.*

Proof. Since $M \mid m_0(r_0)$ and $r_0 \leq X$, one has $M \in S_X$. The first owner itself is not hired at this window: nothing smaller than r_0 can have hired it, so $r_0 \notin S_X$. As a 2-power door, M has gold out-degree zero. No other owner of M exists below r_0 , so no incoming gold arc is live either. Thus M meets no other odd hired prime by a gold edge. $\square$

Every known Mersenne prime from M_{31} onward has first owner larger than the connecting window $X^* = 92\,274\,421$ of the hire-graph note, so Lemma 1 already supplies many concrete disconnected windows. That is a finite list, not an infinitude theorem.

3 Downward chains

Lemma 2. *Let S be a finite nonempty set of odd primes other than 3, closed under taking odd prime factors of m_0 . Then S contains a 2-power door.*

Proof. Take a least element $p \in S$. If $m_0(p) = 2^k$, we are done. Otherwise some odd prime $q \neq 3$ divides $m_0(p)$. Closure puts $q \in S$, and the size bound $q \leq m_0(p)/2 < (p+1)/2$ forces $q < p$, contradicting minimality. $\square$

Gold arcs on odd leaves are strictly decreasing in the prime label, so every finite descending gold chain ends at a sink, and Lemma 2 says every such sink in a door-closed set is a 2-power door.

4 The equivalence

Theorem 1. *The following are equivalent.*

1. *There are infinitely many finite windows X on which undirected gold is disconnected.*
2. *There are infinitely many 2-power-door primes.*
3. *There are infinitely many Mersenne primes, or infinitely many Fermat primes.*

Proof. Lemma 1 gives $(2) \Rightarrow (1)$: each sufficiently large 2-power door contributes at least its first-owner window. For $(1) \Rightarrow (2)$, suppose only finitely many 2-power doors exist. After the last of them has been absorbed into the infinite component of 5 (which the hire-graph note supplies for every odd prime other than 3), every later hired vertex has a downward gold path onto an already-absorbed sink. That path is live in the window: $S_X \setminus \{2\}$ is always closed under odd prime factors of m_0 , because every odd vertex is $\leq (X+1)/2 \leq X$ and is therefore an owner, so its odd door factors already sit in S_X . Hence later windows stay connected. The equivalence of (2) and (3) is the classification of 2-power doors as Mersenne or Fermat primes greater than 3. $\square$

Fermat primes are widely believed finite (five known; the last is 65537). Infinitude of Mersenne primes is open. Thus Theorem 1 does not settle item (1) unconditionally; it identifies the obstruction.

Remark 1 (Heuristic). *The Lenstra–Pomerance–Wagstaff heuristic predicts about $e^\gamma \log_2 \log_2 x$ Mersenne primes with exponent at most x , hence infinitely many, and therefore infinitely many disconnected windows, with a new island of length comparable to M_p around each large Mersenne prime. That is a labelled heuristic, not a proof of (1).*

5 The M_{31} integers

Let $M_{31} = 2^{31} - 1$. A thin CRT hunt (not a gold-graph rebuild) finds:

$$r_0 = 98\,784\,247\,763, \quad v = 300\,647\,710\,579,$$

where r_0 is the first owner of M_{31} and v is the least prime in either class modulo $15M_{31}$ with $5M_{31} \mid m_0(v)$ (class A; $m_0(v) = 2^2 \cdot 5 \cdot 7 \cdot M_{31}$). The ratio $v/r_0 \approx 3.04$. The floor candidate $2M_{31} - 1$ is composite; no earlier owner appears in the two classes modulo $3M_{31}$.

At $X = v$ the integer v is an owner, so 5 and M_{31} divide its door, but $v \notin S_X$, and the gold arcs $v \rightarrow 5$ and $v \rightarrow M_{31}$ are not live until v itself is hired. That hiring time is at least $2v - 1 \approx 6.01 \cdot 10^{11}$. What v records is a bridge owner, not yet a bridge vertex. A mixed owner hired earlier than v could still connect M_{31} sooner; “first direct 5– M_{31} bridge” is not automatically the first connecting vertex. A full component check at these scales remains past home computing.

Lean

Lemmas 1–2 and Theorem 1 are checked in Lean 4 as `Hire/GoldDisconnects.lean` (`goldIsolated_of_firstOwnerExists_twoPowerDoor_of_finite_closed_infinite_goldDisconnected_of_infinite_mersenne_or_fermat`). The M_{31} integers above are thin machine searches, not Lean theorems.

References

- [1] J. Pickett, *The hire graph of the 3-free door*, note, 19th September 2026.